

COMSATS UNIVERSITY ISLAMABAD, WAH CAMPUS

WEB TECHNOLOGIES • LAB 3 — CSS

Lab Report

FOOD CORNER

Integrated CSS Website Development

“A single-page restaurant website built with HTML and CSS”

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Lab	Lab 3 — CSS
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Folder	Lab 3 / FoodCorner v2

CLO-4: Implement functional web applications using appropriate front-end and back-end technologies.

CLO-5: Develop functional web applications using appropriate development technologies.

1 Objective

The task was to develop a single-page website for a fictional restaurant called **Food Corner**, applying every CSS concept covered in the lab: CSS types, selectors, pseudo-classes, pseudo-elements, cascade, specificity, inheritance, colors, fonts, text and background properties, the box model, margin collapsing, width/height constraints, overflow, display properties, borders and shadows. The website is built with a fresh “sticker-style bistro” design: deep green, coral and sun-yellow palette, fat uppercase headings, pill buttons and hard offset shadows.

2 Files Created

```
FoodCorner v2 /  
  index.html          single-page website (structure + demos)  
  style.css           external stylesheet (all main styling)  
  images /  
    pizza.png, burger.png, sandwich.png, pasta.png  
    bg-pattern.png    tileable pattern for the hero section  
  Lab Report.pdf      this report
```

3 Webpage Structure

- **Header** — brand “FOOD.CORNER”, open-hours sticker, navigation: Home, Menu, About, Contact
- **Welcome Section** (hero) — kicker line, main heading “Taste the good stuff”, short paragraph, Explore Menu button
- **Menu Section** — 4 food cards (Pizza, Burger, Sandwich, Pasta); each card has a food image, name, description, rotated price sticker and an Order Now button
- **About Section** — heading, paragraph, plus a rotated Quick-facts card
- **Contact Section** — form with Name, Email, Message and a Submit (“Send it”) button
- **CSS Lab Demos** — five numbered boxes: margin collapsing, constraints, overflow, display, cascade & specificity
- **Footer** — brand line and © 2026 Food Corner

4 CSS Types Used (all three)

Type	Where it is used	Why it is placed there
External CSS	style.css, linked from the head of index.html	The main stylesheet that styles the whole website
Internal CSS	<style> block in the head of index.html	Page-specific rules for the top ticker bar only
Inline CSS	style attribute on the "OPEN TILL 11 PM" badge	One-off sticker style that belongs to exactly one element

5 Selectors Used (all six)

Selector	Example from style.css	What it targets
Element	h2 { line-height: 1.15; }	Every h2 heading on the page
Class	.card { background-color: #fff; }	All four food cards
ID	#home { text-align: center; }	Welcome section (id="home")
Attribute	a[href^="#"] { text-decoration: none; }	Links whose href starts with #
Grouping	h1, h2, h3 { font-family: "Arial Black"; }	All headings in one rule
Descendant	nav ul li a { color: #fff7ec; }	Links inside the navigation list

6 Pseudo-Classes and Pseudo-Elements

Pseudo-classes (:hover and :first-child)

- **nav ul li a:hover** — navigation pills flip to a sun-yellow background with underlined text
- **.order-btn:hover** — Order Now buttons turn deep green, lift up and grow a hard shadow
- **.cards .card:first-child** — the first food card (Pizza) gets a coral border, coral shadow and a small rotation, plus a coral price sticker

Pseudo-elements (::before and ::after)

- **::before** on every .section-title — inserts a fork-and-knife symbol before the heading
- **::after** on every .section-title — draws a coral swash bar under the heading (content empty, styled as a block)

7 Cascade, Specificity and Inheritance

Inheritance: the body sets font-family: "Segoe UI"... and color: #14261f. Every paragraph and most elements inherit these values automatically without repeating the rules.

Overriding inheritance: the footer sets its own color: #cfe0d8 and the hero sets color: white, so both ignore body's dark text colour.

Conflict 1 — different specificity (the more specific selector wins):

```
.card { color: #8a8f8c; }          <- specificity 0-1-0
.menu-section .card { color: #14261f; } <- specificity 0-2-0, WINS
```

The menu cards therefore keep the dark ink colour, not the grey.

Conflict 2 — same specificity (the cascade / source order wins):

```
.specificity-demo { background-color: red; }    <- written first, loses
.specificity-demo { background-color: green; }  <- written later, WINS
```

Both rules have equal weight, so the later rule wins and the demo box on the page appears green.

8 Margin Collapsing

What margin collapsing means

When two block-level elements are stacked vertically and the bottom margin of the first element meets the top margin of the second element, the browser does not add the two margins together. Instead the margins **merge (collapse)** into a single gap equal to the **larger** of the two values.

Demonstration on the page: Box A has margin-bottom: 40px and Box B has margin-top: 20px. The visible gap between them measures **40px** — not $40 + 20 = 60\text{px}$ — proving the smaller margin was absorbed by the larger one. Both boxes are shown in Demo 1 of the CSS Lab Demos section.

9 Colors, Fonts, Text and Backgrounds

- **Named colors:** red and green in the specificity demo, white for headings on dark sections
- **HEX colors:** #0b3e35 (header/hero), #ff5e3a (coral accents), #ffc53d (sun yellow), #fff7ec (page background), #14261f (ink text)
- **RGB color:** rgb(214, 233, 226) for the hero paragraph and rgb(64, 51, 45) for the about paragraph
- **Fonts:** font-family (Segoe UI body / Arial Black display), font-size, font-weight (bold), font-style (italic in the hero text and section subtitles)
- **Text:** text-align (center / justify), text-decoration (none on nav links, underline on hover, dotted underline on subtitles), line-height (1.15–1.9)
- **Background:** the hero uses all five properties together — background-color, background-image (bg-pattern.png), background-repeat, background-position and background-size

10 Box Model on the Food Cards

Layer	Value used on .card
Content	food image + name + description + price sticker + button (width: 250px, min-height: 375px)
Padding	padding: 0 0 20px 0 — space between the content and the border
Border	border: 2px solid #14261f with border-radius: 22px (rounded sticker corners)
Margin	margin: 6px outside each card, plus a 34px gap in the flex row

box-sizing: border-box; is set on the card, so padding and border are counted inside the 250px width and all four cards stay exactly the same size.

Each card also uses **border**, **border-radius** and a hard **box-shadow** (sticker look) — satisfying the borders, shadows and spacing requirement, with consistent section padding and card sizes across the page.

11 Width/Height Constraints and Overflow

Constraints box (Demo 2): width: 60%, height: 120px, min-width: 300px, max-width: 700px, min-height: 80px, max-height: 200px. Resizing the browser shows the box grow and shrink until the min/max limits stop it.

Overflow (Demo 3): three identical boxes with text that does not fit:

- **overflow: visible** — text spills outside the box
- **overflow: hidden** — text is cut off at the edge
- **overflow: scroll** — scrollbars appear so the rest can be read

12 Display Properties

- **display: block** — form labels and inputs, the demo strip
- **display: inline** — badge flowing inside the demo paragraph
- **display: inline-block** — navigation links, nav list items, all buttons and the cards
- **display: none** — the hidden “Closed on Sundays” note and a demo span (Demo 4)

13 Final Look

The finished page uses one colour system (deep green / coral / sun yellow) with consistent section padding, equal card sizes, centred headings with matching swash underlines, hard sticker shadows and generous white space. Text is large enough to read comfortably, sections are clearly separated by background colour, and the whole page adapts to small screens through a media query.

14 Requirements Checklist

Requirement	Status	Where
All three CSS types	OK	style.css + <style> ticker + inline hours sticker
All six selectors	OK	style.css section 2 (element / class / ID / attribute / grouping / descendant)
:hover and :first-child	OK	nav pills, Order Now buttons, first pizza card
::before and ::after	OK	every .section-title heading
Cascade, specificity, inheritance	OK	body inheritance, footer/hero overrides, .card vs .menu-section .card, red/green demo box
Named, HEX and RGB colors	OK	red/green/white, #0b3e35 & #ff5e3a, rgb(214,233,226)
Background (5 properties)	OK	hero section with bg-pattern.png
Box model + box-sizing	OK	.card rules (width/padding/border/margin)
Margin collapsing	OK	Box A / Box B in Demo 1
min/max width & height	OK	.constrained-box in Demo 2
Overflow: visible/hidden/scroll	OK	three boxes in Demo 3
Display: block/inline/inline-block/none	OK	form, demo strip, nav, hidden note
Borders, radius, shadows, spacing	OK	cards, form, demo boxes, consistent padding
© 2026 Food Corner footer	OK	index.html footer

15 Live Demo Submission

Live Website Link

(to be filled after deploying to GitHub Pages / Netlify)

16 Declaration

This website and this report are my own work. I can explain and modify every CSS rule used in this project during the live demonstration.

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